Modern knowledge of the red cell plasma membrane and its membrane skeleton began with Marchesi and Steers's identification of spectrin in 1968. Prior to that year, almost the only thing known about membranes was that they contained a lipid bilayer. Indeed, there was a period in the 1960s when it was believed that red cell membranes contained only a single 22.5 kDa protein called "structural protein".

Three major factors can influence fluidity of a cell membrane :

1. Temperature

It will affect the mode of movement and closeness of phospholipids. During cold conditions, they are found together but when it is hot they move away farther from each other.

2. Cholesterol

These molecules are randomly or arbitrarily distributed across the lipid bilayer, so that the layer can be fluid in various environments. Cholesterol clamps phospholipids thereby controlling the unwarranted movement of molecules across.

3. Saturated and unsaturated fatty acids

Fatty acids make the tails of the phospholipid . (i) Saturated fatty acids : they are carbon chains with single bonds between them which makes them straight enabling tight packing. (ii) unsaturated fats are carbon chains with double bonds between them. These double bonds generate bends in the chain, that do not encourage tight packing. This has role in fluidity of the membrane because the twists or bends will increase the space in between the phospholipids. This will eventually leave them not to get frozen at lower temperatures. Further, molecules such as CO_2 and O_2, do need smaller spaces between the phospholipids through which they can move quicky with ease.The two possible bends that can occur are :

1. Cis-unsaturated fats, both sides of the chain stay on the same side
2. Trans-unsaturated fats, both sides of the chain are opposite from each other

1.3.MEMBRANE TRANSPORT

For the maintenance of the integrity of a cell, the internal concentration of various substances should be maintained. The movement of ions, small biomolecules and nutrients should be regulated, while waste products such as carbon dioxide (CO_2) should leave the cells. The membranes are equipped with a special feature that aid in regulation of concentration of the substances inside a cell. The lipid bilayer structure so far discussed does the major part of control transport processes. Generally, the movement of substances across the membranes are categorized based on whether or not energy is required.

i. Passive transport - the movement of substances across the membrane without the expenditure of cellular energy.

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